MAT B42 Cheatsheet

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1 Vector Valued Function

1.1 Vector Field

$\mathbf{F} : A \subset \mathbb{R}^n \to \mathbb{R}^n$

1.2 Flow Lines

If $\mathbf{F}$ is a vector field, a flow line for $\mathbf{F}$ is a path $c(t)$ such that

$$c'(t) = \mathbf{F}(c(t))$$

2 Divergence and Curl

2.1 Divergence

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$$

2.2 Curl

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$$

2.3 Properties of div and curl

Curl of a Gradient

$$\nabla \times (\nabla f) = \mathbf{0}$$

Proof.

$$\begin{aligned}
\nabla \times (\nabla f) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} \\
&= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}\right)\mathbf{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z}\right)\mathbf{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x}\right)\mathbf{k} \\
&= \mathbf{0}
\end{aligned}$$

□

Divergence of a Curl

$$\nabla \cdot (\nabla \times \mathrm{F}) = 0$$

Properties of Vector Field

$$\mathbf{F} = \nabla f \Rightarrow \text{curl}\, \mathbf{F} = \mathbf{0}$$
$$\mathbf{F} = \nabla \times \mathbf{G} \Rightarrow \text{div}\, \mathbf{F} = 0$$

Proof. Let

$$G_1 = \int_0^z F_2(x, y, t)dt - \int_0^y F_3(x, t, 0)dt,\ G_2 \ = -\int_0^z F_1(x, y, t)dt,\ G_3 = 0$$

□

Basic Identities of Vector Analysis

1. $\nabla(f+g) = \nabla f + \nabla g$
2. $\nabla(cf) = c\nabla f$, for a constant c
3. $\nabla(fg) = f\nabla g + g\nabla f$
4. $\nabla(f/g) = (g\nabla f - f\nabla g)/g^2$, at points x where $g(\mathrm{x}) \neq 0$
5. $\operatorname{div}(\mathbf{F}+\mathbf{G}) = \operatorname{div}\mathbf{F} + \operatorname{div}\mathbf{G}$
6. $\operatorname{curl}(\mathbf{F}+\mathbf{G}) = \operatorname{curl}\mathbf{F} + \operatorname{curl}\mathbf{G}$
7. $\operatorname{div}(f\mathbf{F}) = f\operatorname{div}\mathbf{F} + \mathbf{F}\cdot\nabla f$
8. $\operatorname{div}(\mathbf{F}\times\mathbf{G}) = \mathbf{G}\cdot\operatorname{curl}\mathbf{F} - \mathbf{F}\cdot\operatorname{curl}\mathbf{G}$
9. $\operatorname{div}\operatorname{curl}\mathbf{F} = \mathbf{0}$
10. $\operatorname{curl}(f\mathbf{F}) = f\operatorname{curl}\mathbf{F} + \nabla f\times\mathbf{F}$
11. $\operatorname{curl}\nabla f = \mathbf{0}$
12. $\nabla^2(fg) = f\nabla^2 g + g\nabla^2 f + 2(\nabla f\cdot\nabla g)$
13. $\operatorname{div}(\nabla f\times\nabla g) = \mathbf{0}$
14. $\operatorname{div}(f\nabla g - g\nabla f) = f\nabla^2 g - g\nabla^2 f$

3 Intergrals Over Path and Surfaces

3.1 The Path Intergrals

$$\int_{\mathbf{c}} f ds = \int_a^b f(\mathbf{c}(t)) \left\|\mathbf{c}'(t)\right\| dt$$

3.2 Line Integral

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t) dt$$

Theorem 1 Change of Parametrization for Line Integrals
Let $\mathbf{F}$ be a vector field continuous on the C^1 path $\mathbf{c} : [a_1, b_1] \to \mathbb{R}^3$, and let $\mathbf{p} : [a, b] \to \mathbb{R}^3$ be a reparametrization of $\mathbf{c}$. If $\mathbf{p}$ is orientation-preserving, then

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$$

and if $\mathbf{p}$ is orientation-reversing, then

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = -\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$$

Theorem 3 Line Integrals of Gradient Vector Fields
Suppose $f : \mathbb{R}^3 \to \mathbb{R}$ is of class C^1 and that $c : [a, b] \to \mathbb{R}^3$ is a piecewise C^1 path. Then

$$\int_c \nabla f \cdot d\mathbf{s} = f(\mathbf{c}(b)) - f(\mathbf{c}(a))$$

Line Integrals Over Curves with Opposite Orientations
Let C^-be the same curve as C, but with the opposite orientation. Then

$$\int_C \mathbf{F} \cdot d\mathbf{s} = -\int_{C^-} \mathbf{F} \cdot d\mathbf{s}$$

Line Integrals Over Curves Consisting of Several Components
Let C be an oriented curve such that $C = C_1 + C_2 + \cdots + C_k$. So we have:

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{C_1} \mathbf{F} \cdot d\mathbf{s} + \int_{C_2} \mathbf{F} \cdot d\mathbf{s} + \cdots + \int_{C_k} \mathbf{F} \cdot d\mathbf{s}$$

3.3 Area of a Surface

Definition Regular Surface
A surface is regular iff. $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$

Area of a Parametrized Surface

$$A(S) = \iint_D \|\mathbf{T}_u \times \mathbf{T}_v\| \, dudv$$

If $\mathbf{\Phi} = (u, v, g(u, v))$,

$$A(S) = \iint_D \left(\sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1} \right) dA$$

Proof.

$$\mathbf{T}_u = \mathbf{i} + \frac{\partial g}{\partial u}\mathbf{k}, \quad \mathbf{T}_v = \mathbf{j} + \frac{\partial g}{\partial v}\mathbf{k}$$

$$\mathbf{T}_u \times \mathbf{T}_v = -\frac{\partial g}{\partial u}\mathbf{i} - \frac{\partial g}{\partial v}\mathbf{j} + \mathbf{k} = -\frac{\partial g}{\partial x}\mathbf{i} - \frac{\partial g}{\partial y}\mathbf{j} + \mathbf{k}$$

$$||\mathbf{T}_u \times \mathbf{T}_v|| = \sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1}$$

□

Surface of Revolution
The lateral surface area generated by revolving the graph of a function $y = f(x)$ about the x-axis is given by

$$A = 2\pi \int_a^b \left(|f(x)| \sqrt{1 + [f'(x)]^2} \right) dx$$

Proof. Let $S : \mathbf{\Phi}(u, v) = (u, f(u)\cos v, f(u)\sin v)$, so we have:

$$\frac{\partial(x, y)}{\partial(u, v)} = -f(u)\sin v, \quad \frac{\partial(y, z)}{\partial(u, v)} = f(u)f'(u), \quad \frac{\partial(x, z)}{\partial(u, v)} = f(u)\cos v$$

$$\begin{aligned}
A(S) &= \iint_D \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2} dudv \\
&= \iint_D \sqrt{[f(u)]^2 \sin^2 v + [f(u)]^2 [f'(u)]^2 + [f(u)]^2 \cos^2 v} dudv \\
&= \iint_D |f(u)| \sqrt{1 + [f'(u)]^2} dudv \\
&= \int_a^b \int_0^{2\pi} |f(u)| \sqrt{1 + [f'(u)]^2} dvdu \\
&= 2\pi \int_a^b |f(u)| \sqrt{1 + [f'(u)]^2} du
\end{aligned}$$

□

3.4 Surface Integral

1. Parametrized Surface: $\mathbf{\Phi}(u, v)$

   a. Integral of a scalar function f :

$$\iint_S f dS = \iint_D f(\mathbf{\Phi}(u, v)) \left\| \mathbf{T}_u \times \mathbf{T}_v \right\| dudv$$

   b. Scalar surface element:

$$dS = \left\| \mathbf{T}_u \times \mathbf{T}_v \right\| dudv$$

   c. Integral of a vector field $\mathbf{F}$:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) \, dudv$$

   d. Vector surface element:

$$d\mathbf{S} = (\mathbf{T}_u \times \mathbf{T}_v) \, dudv = \mathbf{n} dS$$

2. Graph: $z = g(x, y)$

   a. Integral of a scalar function f :

$$\iint_S f dS = \iint_D \frac{f(x, y, g(x, y))}{\cos\theta} dxdy = \iint_D \frac{f(x, y, g(x, y))}{\mathbf{n} \cdot \mathbf{k}} dxdy$$

   b. Scalar surface element:

$$dS = \frac{dxdy}{\cos\theta} = \sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1} dxdy$$

   where $\cos\theta = \mathbf{n} \cdot \mathbf{k}$, and $\mathbf{n}$ is a unit normal vector to the surface.

   c. Integral of a vector field F :

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left(-F_1 \frac{\partial g}{\partial x} - F_2 \frac{\partial g}{\partial y} + F_3 \right) dxdy$$

   d. Vector surface element:

$$d\mathbf{S} = \mathbf{n} \cdot dS = \left(-\frac{\partial g}{\partial x} \mathbf{i} - \frac{\partial g}{\partial y} \mathbf{j} + \mathbf{k} \right) dxdy$$

3. Sphere: $x^2 + y^2 + z^2 = R^2$

   a. Scalar surface element:

$$dS = R^2 \sin\phi d\phi d\theta$$

   b. Vector surface element:

$$d\mathbf{S} = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) R \sin\phi d\phi d\theta = \mathrm{r} R \sin\phi d\phi d\theta = \mathrm{n} R^2 \sin\phi d\phi d\theta$$

4 Classic Theorems of Vector Calculus

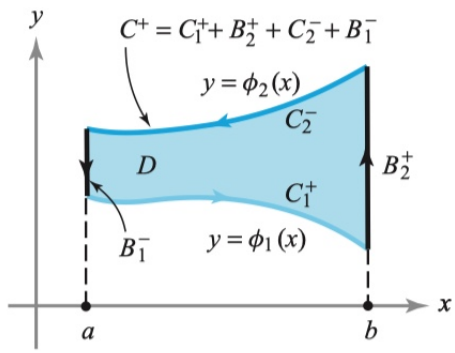
4.1 Green's Theorem

Green's Theorem
If $F = P\mathbf{i} + Q\mathbf{j}$, and C^+ is a simple closed curve(orients anitclockwisly about the region D):

$$\int_{C^+} Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy$$

Proof. **Lemma1**

$$\iint_D \frac{\partial P}{\partial y}(x,y)dxdy = -\int_{C^+} Pdx$$



Proof

$$\begin{aligned}
\iint_D \frac{\partial P}{\partial y}(x,y)dxdy &= \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \frac{\partial P}{\partial y}(x,y)dydx \\
&= \int_a^b \left[P\left(x,\phi_2(x)\right) - P\left(x,\phi_1(x)\right)\right] dx \\
&= -(\int_{C_1^+} Pdx - \int_{C_2^+} Pdx) \\
&= -(\int_{C_1^+} Pdx + 0 + \int_{C_2^-} Pdx + 0) \\
&= -(\int_{C_1^+} Pdx + \int_{B_2^+} Pdx + \int_{C_2^-} Pdx + \int_{B_1^-} Pdx) \\
&= -\int_{C^+} Pdx
\end{aligned}$$

QED.

Lemma 2

$$\int_{C^+} Qdy = \iint_D \frac{\partial Q}{\partial x} dxdy$$

PROOF

$$\iint_D \frac{\partial Q}{\partial x} dxdy = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} \frac{\partial Q}{\partial x} dxdy = \int_c^d \left[Q\left(\psi_2(y), y\right) - Q\left(\psi_1(y), y\right)\right] dy$$

$$= \int_{C_2^+} Qdy - \int_{C_1^+} Qdy = \int_{C_2^+} Qdy + \int_{C_1^-} Qdy = \int_{C^+} Qdy$$

QED.

By combining Lemma 1 and Lemma 2, we thus proved Green's Theorem. □

NOTE:
For non-simple region, we can divides the region into sub-regions and apply Green's Theorem.

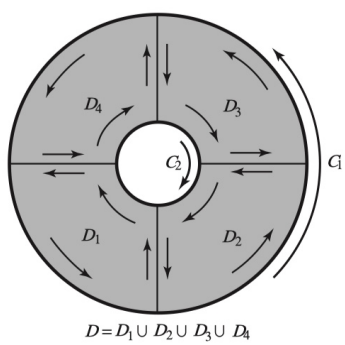


figure **8.1.5** Green's theorem applies to $D = D_1 \cup D_2 \cup D_3 \cup D_4$.

Area of a Region
Can be proved by Green's Theorem

$$A = \frac{1}{2} \int_{\partial D} xdy - ydx$$

4.2 Stoke's Theorem

Theorem 6 Stoke's Theorem
If S is a C^2 surface parametrized by $\mathbf{\Phi}(u,v) = (u, v, f(u,v))$, and $\mathbf{F}$ is a C^1 vector field on S, ∂S is the boundary of S, we have

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}$$

Proof.

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_D \left[\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \left(-\frac{\partial z}{\partial x} \right) + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \left(-\frac{\partial z}{\partial y} \right) + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \right] dA$$

And,

$$\begin{aligned}
\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} &= \int_a^b \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt \\
&= \int_a^b \left[\left(F_1 + F_3 \frac{\partial z}{\partial x} \right) \frac{dx}{dt} + \left(F_2 + F_3 \frac{\partial z}{\partial y} \right) \frac{dy}{dt} \right] dt \\
&= \int_{\mathbf{c}} \left(F_1 + F_3 \frac{\partial z}{\partial x} \right) dx + \left(F_2 + F_3 \frac{\partial z}{\partial y} \right) dy \\
&= \int_{\partial D} \left(F_1 + F_3 \frac{\partial z}{\partial x} \right) dx + \left(F_2 + F_3 \frac{\partial z}{\partial y} \right) dy.
\end{aligned}$$

By applying Green's Theorem, we have:

$$\begin{aligned}
\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} &= \iint_D \left[\frac{\partial \left(F_2 + F_3 \partial z / \partial y \right)}{\partial x} - \frac{\partial \left(F_1 + F_3 \partial z / \partial x \right)}{\partial y} \right] dA \\
&= \iint_D \left[\left(\frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial z} \frac{\partial z}{\partial x} + \frac{\partial F_3}{\partial x} \frac{\partial z}{\partial y} + \frac{\partial F_3}{\partial z} \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} + F_3 \frac{\partial^2 z}{\partial x \partial y} \right) \right. \\
&\quad \left. - \left(\frac{\partial F_1}{\partial y} + \frac{\partial F_1}{\partial z} \frac{\partial z}{\partial y} + \frac{\partial F_3}{\partial y} \frac{\partial z}{\partial x} + \frac{\partial F_3}{\partial z} \frac{\partial z}{\partial y} \frac{\partial z}{\partial x} + F_3 \frac{\partial^2 z}{\partial y \partial x} \right) \right] dA \\
&= \iint_D \left[\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \left(-\frac{\partial z}{\partial x} \right) + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \left(-\frac{\partial z}{\partial y} \right) + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \right] dA \\
&= \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S}
\end{aligned}$$

□

4.3 Conservative Fields

Theorem 7 Conservative Fields
Statements below are equivalent:

(i) For any oriented simple closed curve C, $\int_C \mathbf{F} \cdot d\mathbf{s} = 0$.

(ii) For any two oriented simple curves C_1 and C_2 that have the same endpoints,

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{s} = \int_{C_2} \mathrm{F} \cdot d\mathbf{s}$$

(iii) $\exists f \ s.t. \ \mathbf{F} = \nabla f$
(and if $\mathbf{F}$ has one or more exceptional points where it fails to be defined, f is also undefined there).

(iv) $\nabla \times \mathbf{F} = \mathbf{0}$.

Proof. In order to prove the theorem, we establish following chain of implication:

$$(i) \to (ii) \to (iii) \to (iv) \to (i)$$

Proof of $(i) \to (ii)$:
Let $\mathbf{c} = \mathbf{c_1} - \mathbf{c_2}$. By (i):

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}_1} \mathbf{F} \cdot d\mathbf{s} - \int_{\mathbf{c}_2} \mathbf{F} \cdot d\mathbf{s} = 0$$

$$\leftrightarrow \int_{C_1} \mathbf{F} \cdot d\mathbf{s} = \int_{C_2} \mathbf{F} \cdot d\mathbf{s}$$

QED.

Proof of $(ii) \to (iii)$:
From (ii), we know that $\mathbf{F}$ is path independent, so we can set up different path to complete the proof. Here's a example for proving $\frac{\partial f}{\partial z} = F_3$
Choose C is a simple curve with the path (0, 0, 0) to (x, 0, 0) to (x, y, 0) to (x, y, z)
Define, $f = \int_0^x F_1(t, 0, 0)dt + \int_0^y F_2(x, t, 0)dt + \int_0^z F_3(x, y, t)dt$. Therefore, $\frac{\partial f}{\partial z} = F_3$.

QED.

Proof of $(iii) \to (iv)$:
See in section2.3

Proof of $(iv) \to (i)$:

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = \iint_S \mathbf{0} \cdot \mathbf{n} dS = 0$$

QED.

□

Corollary 1
F is a C^1 vector field on $\mathbb{R}^2$ of the form $P\mathbf{i} + Q\mathrm{j}$ that satisfies $\partial P/\partial y = \partial Q/\partial x$, then $\mathrm{F} = \nabla f$ for some f on $\mathbb{R}^2$.

Theorem 8
If F is a C^1 vector field on all of $\mathbb{R}^3$ with $\operatorname{div}\mathbf{F} = 0$, then there exists a C^1 vector field $\mathbf{G}$ with $\mathbf{F} = \operatorname{curl}\mathbf{G}$.

Proof. Define $\mathbf{G} = G_1\mathbf{i} + G_2\mathbf{j} + G_3\mathbf{k}$ by

$$\begin{aligned}
G_1(x,y,z) &= \int_0^z F_2(x,y,t)dt - \int_0^y F_3(x,t,0)dt \\
G_2(x,y,z) &= -\int_0^z F_1(x,y,t)dt \\
G_3(x,y,z) &= 0
\end{aligned}$$

Therefore, $\nabla \times \mathbf{G} = (F_1, F_2, F_3) = \mathbf{F}$ □

4.4 Gauss Theorem

Theorem 9 Gauss' Divergence Theorem

$$\iiint_W (\operatorname{div}\mathbf{F})dV = \iiint_W (\nabla\cdot\mathbf{F})dV = \iint_{\partial W} \mathbf{F}\cdot d\mathbf{S}$$

Theorem 10 Gauss' Law

$$\iint_{\partial M} \frac{\mathbf{r}\cdot\mathbf{n}}{||\mathbf{r}||^3}dS = \begin{cases} 4\pi & \text{if } (0,0,0)\in M \\ 0 & \text{if } (0,0,0)\notin M \end{cases}$$

Proof. If $(0,0,0)\notin M$. Then $\frac{\mathbf{r}}{r^3}$ is C^1. So by Gauss' Theorem,

$$\iint_{\partial W} \frac{\mathbf{r}\cdot\mathbf{n}}{||\mathbf{r}||^3}dS = \iiint_W \operatorname{div}\left(\frac{\mathbf{r}}{||\mathbf{r}||^3}\right)dV = \iiint_W 0dV = 0$$

If $(0,0,0)\in M$. Because M is no longer C^1, so we needs a unit ball N centered $(0,0,0)$ of radius $\epsilon > 0$ to exclude the origin. Therefore,

$$\begin{aligned}
\iint_{\partial N} \mathbf{F}\cdot\mathbf{n}dS &= \iint_{\partial N} \frac{(x,y,z)\cdot\left(\frac{x}{\epsilon},\frac{y}{\epsilon},\frac{z}{\epsilon}\right)}{(x^2+y^2+z^2)^{3/2}}dS \\
&= \frac{1}{\epsilon}\iint_{\partial N} \frac{x^2+y^2+z^2}{(x^2+y^2+z^2)^{3/2}}dS = \frac{1}{\epsilon}\iint_{\partial N} \frac{1}{\sqrt{x^2+y^2+z^2}}dS \\
&= \frac{1}{\epsilon}\iint_{\partial N} \frac{1}{\sqrt{\epsilon^2}}dS = \frac{1}{\epsilon^2}\iint_{\partial N} dS = \frac{1}{\epsilon^2}\left(4\pi\epsilon^2\right) = 4\pi
\end{aligned}$$

As the result,

$$\iint_{\partial W} \frac{\mathbf{r}\cdot\mathbf{n}}{||\mathbf{r}||^3} = \iint_{\partial N} \mathbf{F}\cdot\mathbf{n}dS + \iint_{\partial W\setminus\partial N} \mathbf{F}\cdot\mathbf{n}dS = 4\pi$$

□

4.5 Differential form

Definition Differentiation $d : \bigwedge^k \to \bigwedge^{k+1}$, with properties:

1. If $f \in \bigwedge^0$, then $df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial x}dz$
2. $d(\omega_1 + \omega_2) = d\omega_1 + d\omega_2$
3. If $\omega \in \bigwedge^k$, then $d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^k \omega \wedge (d\eta)$
4. As a linear map $d^2 = 0$

Definition Wedge $\wedge : \bigwedge^k \times \bigwedge^\ell \to \bigwedge^{k+\ell}$, with porperties:

1. $0 \wedge \omega = 0$
2. $(f\omega_1 + \omega_2) \wedge \eta = (f)(\omega_1 \wedge \eta) + \omega_2 \wedge \eta$
3. $\omega \wedge \eta = (-1)^{k\ell} \eta \wedge \omega$
4. $\omega_1 \wedge (\omega_2 \wedge \omega_3) = (\omega_1 \wedge \omega_2) \wedge \omega_3$
5. If f is a 0 -form then $f \wedge \omega = f\omega$

Definition
A form is closed $d\omega = 0$ and it is exact if $\omega = d\eta$.
Theorem
Exact implies closed.

Theorem 11 Green's Theorem
D is an elementary region, with ∂D given the counterclockwise orientation. Suppose $\omega = P(x,y)dx + Q(x,y)dy$ is a 1-form on some open set K in $\mathbb{R}^3$ that contains D. Then

$$\int_{\partial D} \omega = \iint_D d\omega$$

Theorem 12 Stokes' Theorem
S is an oriented surface on $\mathbb{R}^3$ with boundary ∂S.Suppose ω is a 1-form on some open set K that contains S. Then

$$\int_{\partial S} \omega = \iint_S d\omega$$

Theorem 13 Gauss' Theorem
Let $W \subset \mathbb{R}^3$ be an elementary region with ∂W given the outward orientation. If η is a 2-form on some region K containing W, then

$$\iint_{\partial W} \eta = \iiint_W d\eta$$

5 Fourier expansion and PDEs

5.1 Fourier polynomial

Definition

$$\langle f(x), g(x)\rangle = \int_{-\pi}^{\pi} f(x)g(x)dx$$

This is a symmetric bilinear inner product. The corresponding norm is L^2-norm:

$$\|f\|_2^2 = \int_{-\pi}^{\pi} [f(x)]^2 dx$$

Lemma
For a inner product space $(V, \langle\cdot,\cdot\rangle)$. If $\{\vec{v}_0, ..., \vec{v}_n\}$ is an orthogonal set of vectors and $\vec{x} \in \text{span}\{\vec{v}_0, ..., \vec{v}_n\}$, then:

$$\vec{x} = \frac{\langle \vec{x}, \vec{v}_0\rangle}{\langle \vec{v}_0, \vec{v}_0\rangle}\vec{v}_0 + \frac{\langle \vec{x}, \vec{v}_1\rangle}{\langle \vec{v}_1, \vec{v}_1\rangle}\vec{v}_1 + \cdots + \frac{\langle \vec{x}, \vec{v}_n\rangle}{\langle \vec{v}_n, \vec{v}_n\rangle}\vec{v}_n$$

Definition
The Fourier basis for $L^2([-\pi, \pi])$ is

$$\{1\} \cup \bigcup_{n\in\mathbb{N}} \{\cos(nx), \sin(nx)\}$$

Definition
The n-th Fourier polynomial $F_n(x)$ of $f(x)$ has the form:

$$F_n(x) = a_0 + \sum_{k=1}^{n} a_k \cos(kx) + \sum_{k=1}^{n} b_k \sin(kx)$$

where:

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx \quad a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(kx)dx \quad b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(kx)dx$$

Proof.

$$a_0 = \frac{\langle 1, f\rangle}{\langle 1, 1\rangle} \quad a_k = \frac{\langle \cos(kx), f\rangle}{\langle \cos(kx), \cos(kx)\rangle} \quad b_k = \frac{\langle \sin(kx), f\rangle}{\langle \sin(kx), \sin(kx)\rangle}$$

as you should verify □

Definition
The energy of $f(x)$ is:

$$E = \frac{1}{\pi}\int_{-\pi}^{\pi} (f(x))^2 dx$$

Alternatively, $E = \frac{1}{\pi}||f||_2^2$

5.2 Partial differential equations

Wave Equation
Wave equations has the format

$$u_{tt} = c^2 u_{xx}$$

It has solution

$$u(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right)\left(A_n \cos\left(\frac{n\pi ct}{l}\right) + B_n \sin\left(\frac{n\pi ct}{l}\right)\right)$$

Where

$$A_n = \frac{2}{l}\int_0^l \phi(x)\sin\left(\frac{n\pi x}{l}\right)dx,\ \ B_n = \frac{2}{n\pi c}\int_0^l \psi(x)\sin\left(\frac{n\pi x}{l}\right)dx$$

Heat equation
Heat equations has the format

$$u_t = k u_{xx}$$

It has solution

$$u(x,t) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{l}\right) e^{-\left(\frac{n\pi}{l}\right)^2 kt}$$

Where

$$A_0 = \frac{1}{l}\int_0^l \phi(x)dx,\ \ A_m = \frac{2}{l}\int_0^l \phi(x)\cos\left(\frac{m\pi x}{l}\right)dx, \quad m \neq 0$$